# Category Theory Course Notes

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## Remark

This is a personal document that is intended to help me with understanding of Sheaves and Cohomology course at Kyiv School of Economics by Pierre Schapira, based on his awesome Lecture Notes "An Introduction to Categories and Homological Algebra".

# 1 Language of Categories

## 1.1 Basic Theory

In what follows we denote by $\mathrm{Hom}(X, Y)$ the set of mappings from $X$ to $Y$.

**Definition 1.1** (Category)**.** A **category** $\mathscr{C}$ consists of objects $\mathrm{ob}(\mathscr{C})$, morphisms $\mathrm{Hom}_{\mathscr{C}}(X, Y)$ for each $X, Y \in \mathrm{ob}(\mathscr{C})$, and associative composition map

$$\circ : \mathrm{Hom}_{\mathscr{C}}(X, Y) \times \mathrm{Hom}_{\mathscr{C}}(Y, Z) \to \mathrm{Hom}_{\mathscr{C}}(X, Z)$$

We additionally require existence of an identity function $\mathrm{id}_X \in \mathrm{Hom}(X, X)$ associated with every $X \in \mathrm{ob}(\mathscr{C})$ such that $f \circ \mathrm{id}_X = f$ and $\mathrm{id}_X \circ g = g$ for arbitrary morphisms $f \in \mathrm{Hom}_{\mathscr{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathscr{C}}(Y, X)$.

**Definition 1.2** (Types of morphisms)**.** We have the following types of morphisms:

- A morphism $f \in \mathrm{Hom}(X, Y)$ is an **isomorphism** if there exists $g \in \mathrm{Hom}(Y, X)$ such that $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$.
- A morphism $f : X \to Y$ is an **monomorphism** (resp. an epimorphism) if for any morphisms $g_1, g_2 : X \rightrightarrows Y$ the condition $f \circ g_1 = f \circ g_2$ (and $g_1 \circ f = g_2 \circ g$ resp.) implies $g_1 = g_2$.

**Definition 1.3.** An object $\emptyset_{\mathscr{C}} \in \mathscr{C}$ is called **initial** if $\mathrm{Hom}(\emptyset_{\mathscr{C}}, X) \cong \{\mathrm{pt}\}$ for all $X \in \mathscr{C}$. An object $\mathrm{pt}_{\mathscr{C}}$ is **terminal** if it is initial in $\mathscr{C}^{\mathrm{op}}$. Object $0_{\mathscr{C}}$ is a **zero-object** if it is both initial and terminal.

**Definition 1.4** (Functor)**.** Let $\mathscr{C}$ and $\mathscr{C}'$ be two categories. A **functor** $F : \mathscr{C} \to \mathscr{C}'$ consists of two maps: (1) mapping between objects $F : \mathrm{ob}(\mathscr{C}) \to \mathrm{ob}(\mathscr{C}')$ and (2) for all $X, Y \in \mathscr{C}$ mapping from one set of morphisms to another $F : \mathrm{Hom}_{\mathscr{C}}(X, Y) \to \mathrm{Hom}_{\mathscr{C}'}(F(X), F(Y))$ such that:

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}, \quad F(f \circ g) = F(f) \circ F(g)$$

**Definition 1.5** (Types of Functor)**.** Let $F : \mathscr{C} \to \mathscr{C}'$ be a functor. Then,

- One says that $F$ is **faithful/full** if for any pair $X, Y \in \mathscr{C}$, the mapping $\mathrm{Hom}_{\mathscr{C}}(X, Y) \to \mathrm{Hom}_{\mathscr{C}}(F(X), F(Y))$ is injective/surjective.
- One says that $F$ is **essentially surjective** if for each $Y \in \mathscr{C}'$ there exists $X \in \mathscr{C}$ such that $F(X) \cong Y$.
- One says that $F$ is **conservative** if a morphism $f : X \to Y$ in $\mathscr{C}$ is an isomorphism as soon as $F(f)$ is an isomorphism.

Now, given two functors $F_1, F_2 : \mathscr{C} \rightrightarrows \mathscr{C}'$, we want to define their equivalence. For that reason, we first need to define how to draw a connection between two functors. This is done by means of the following definition.

**Definition 1.6.** Let $F, G$ be two functors from $\mathscr{C}$ to $\mathscr{C}'$. The **natural transformation** (or morphism of functors) $\alpha$ is the morphism $\alpha(X) : F(X) \to G(X)$, indexed by some object $X \in \mathscr{C}$, such that the for any morphism $f \in \mathrm{Hom}_{\mathscr{C}}(X, Y)$ the diagram below commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\alpha(X)} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow{\alpha(Y)} & G(Y) \end{array}$$

Since we have the notion of morphism between functors, we can form the category of functors. Namely, denote by $\mathrm{Fct}(\mathscr{C}, \mathscr{C}')$ the category of functors from $\mathscr{C}$ to $\mathscr{C}'$.

Now, naturally, we sometimes want to state the equivalence of two categories (say, category of finite-dimensional matrices and finite vector spaces). One "obvious" way to define equivalence is to require the existence of bijective functor, but this notion turns out to be too strict. For that reason, we require the existence of a map that, composed with the functors left and right, is isomorphic to the identity maps.

**Definition 1.7** (Equivalence of categories)**.** A functor $F : \mathscr{C} \to \mathscr{C}'$ is an **equivalence** of categories if there exists $G : \mathscr{C}' \to \mathscr{C}$ such that $G \circ F \cong \mathrm{id}_{\mathscr{C}}$ and $F \circ G \cong \mathrm{id}_{\mathscr{C}'}$.

A slightly easier equivalent to this definition is following.

**Theorem 1.8.** The functor $F : \mathscr{C} \to \mathscr{C}'$ is an equivalence of categories if and only if $F$ is fully faithful and essentially surjective.

Now denote by $\mathscr{C}^\wedge = \mathrm{Fct}(\mathscr{C}^{\mathrm{op}}, \mathbf{Set})$ and $\mathscr{C}^\vee = \mathrm{Fct}(\mathscr{C}^{\mathrm{op}}, \mathbf{Set}^{\mathrm{op}})$. Define the corresponding functors between $\mathscr{C}$ and given two categories above:

$$\begin{aligned} h_{\mathscr{C}} &: \mathscr{C} \to \mathscr{C}^\wedge, \quad X \mapsto \mathrm{Hom}_{\mathscr{C}}(\cdot, X), \\ k_{\mathscr{C}} &: \mathscr{C} \to \mathscr{C}^\vee, \quad X \mapsto \mathrm{Hom}_{\mathscr{C}}(X, \cdot). \end{aligned}$$

Since there is a natural equivalence between $\mathscr{C}^\wedge$ and $\mathscr{C}^\vee$: namely, $\mathscr{C}^\vee \cong \mathscr{C}^{\mathrm{op},\wedge,\mathrm{op}}$, we focus on $\mathscr{C}^\wedge$.

The following Lemma is fundamental in the elementary category theory. In its essence, it asks a natural question: what can we say about $X$, given the set of morphisms to $X$ (that is, $\mathrm{Hom}_{\mathscr{C}}(\cdot, X)$)?

**Lemma 1.9** (The Yonede lemma)**.** For functor $A \in \mathscr{C}^\wedge$ and $X \in \mathscr{C}$ there is an isomorphism $\mathrm{Hom}_{\mathscr{C}^\wedge}(h_{\mathscr{C}}(X), A) \cong A(X)$, functorial with respect to $X$ and $A$.

## 1.2 Problems

**Problem.** In a category $\mathscr{C}$, consider three morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} W$. Prove that if $f \circ g$ and $g \circ h$ are isomorphisms, then $f$ is an isomorphism.

**Proof.**

**Problem.** Prove that the categories **Set** and $\mathbf{Set}^{\mathrm{op}}$ are not equivalent and similarly with the categories $\mathbf{Set}^f$ and $\mathbf{Set}^{f,\mathrm{op}}$.

**Proof.** Assume the contrary: suppose there exists an equivalence functor $F : \mathbf{Set} \to \mathbf{Set}^{\mathrm{op}}$. Then, one obviously has $F(\emptyset) \cong \{\mathrm{pt}\}$ and $F(\{\mathrm{pt}\}) \cong \emptyset$. Recall that $F$ must be fully faithful, meaning for each $X, Y \in \mathscr{C}$ one has:

$$\mathrm{Hom}_{\mathbf{Set}}(X, Y) \xrightarrow{\cong} \mathrm{Hom}_{\mathbf{Set}^{\mathrm{op}}}(F(X), F(Y))$$

Now, take $X := \{\mathrm{pt}\}$ and $Y$ to be any finite set with more than two elements (say, $Y$ is any set of two elements). Then, the left handside $\mathrm{Hom}_{\mathbf{Set}}(\{\mathrm{pt}\}, Y)$ contains more than 1 element. In contrast, the right handside is exactly $\mathrm{Hom}_{\mathbf{Set}}(F(Y), \emptyset)$, which contains exactly 1 element. Obviously, two finite sets cannot be isomorphic when they are of different sizes, so we reached the contradiction.

**Problem.** This problem consists of two parts:

1. Let $F : \mathscr{C} \to \mathscr{C}'$ be a faithful functor and let $f$ be a morphism in $\mathscr{C}$. Prove that if $F(f)$ is a monomorphism (resp. an epimorphism), then $f$ is a monomorphism (resp. an epimorphism).
2. Assume now that $F$ is fully faithful. Prove that if $F(f)$ is an isomorphism, then $f$ is an isomorphism. In other words, fully faithful functors are conservative.

**Proof.**

*Part 1.* We know two facts about $F$ and $f$:

- Since $F$ is a faithful functor, we have $\mathrm{Hom}_{\mathscr{C}}(X, Y) \to \mathrm{Hom}_{\mathscr{C}'}(F(X), F(Y))$ is injective.
- Since $F(f)$ is a monomorphism, by definition for any $g_1, g_2 : X \rightrightarrows Y$ one has the following: $F(f) \circ g_1 = F(f) \circ g_2 \implies g_1 = g_2$.

Now, pick $g_1, g_2 : X \rightrightarrows Y$ and we ought to show that $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$. Apply functors on both sides, so we have:

$$F(f \circ g_1) = F(f \circ g_2) \iff F(f) \circ F(g_1) = F(f) \circ F(g_2)$$

Now since $F(f)$ is a monomorphism, this implies $F(g_1) = F(g_2)$. But since $F$ is faithful, that implies $g_1 = g_2$. Thus, $f$ is a monomorphism.

Epimorphism is proved in the similar manner, but with $g_j \circ f$ instead of $f \circ g_j$.

*Part 2.* Suppose $f \in \mathrm{Hom}_{\mathscr{C}}(X, Y)$. Since $F(f)$ is an isomorphism, there exists a map $g \in \mathrm{Hom}_{\mathscr{C}'}(F(Y), F(X))$ such that $F(f) \circ g = \mathrm{id}_{F(Y)}$ and $g \circ F(f) = \mathrm{id}_{F(X)}$. We now shall construct a map $h \in \mathrm{Hom}_{\mathscr{C}}(Y, X)$ such that $f \circ h = \mathrm{id}_Y$ and $h \circ f = \mathrm{id}_X$. If such map exists, then by applying

functor we arrive at:

$$F(f) \circ F(h) = \mathrm{id}_{F(Y)}, \quad F(h) \circ F(f) = \mathrm{id}_{F(X)}$$

So it motivates us to set $h$ such that $F(h) = g$. Since $F$ is fully faithful, we can set $h := F^{-1}(g)$.

**Problem.** Is the natural functor **Set** $\to$ **Rel** full, faithful, fully faithful, conservative?

**Answer.** We first construct the functor $F : \mathbf{Set} \to \mathbf{Rel}$. For objects, we have $X \xrightarrow{F} X$, while we associate every mapping $f \in \mathrm{Hom}_{\mathbf{Set}}(X, Y)$ with graph $F(f) = \{(x, f(x)) : x \in X\} \subseteq X \times Y$. Just for practicing purposes, let us show that $F$ is indeed a valid functor. First, note that $F(\mathrm{id}_X) = \Delta_{X \times X}$ and for $f \in \mathrm{Hom}_{\mathbf{Rel}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathbf{Rel}}(Y, Z)$ we have:

$$F(f) \circ F(g) = \{(x, z) : \exists y \text{ such that } (x, y) \in F(f), (y, z) \in F(g)\}$$

But note that $(x, y) \in F(f)$ means that $y = f(x)$ and similarly $(y, z) \in F(g)$ means $z = f(y)$ which in turn is nothing but $z = (f \circ g)(x)$. Thus,

$$F(f) \circ F(g) = \{(x, (f \circ g)(x)) : x \in X\} = F(f \circ g)$$

Good, now we can finally check all the properties.

*(1) Fullness.* Is the map surjective? Surely not as most relations are not graphs. For instance, one cannot represent the relation $\{(1, a), (1, b)\}$ for $a \neq b$ using $F$.

*(2) Faithfulness.* Is the map injective? Suppose $F(g_1) = F(g_2)$, meaning

$$\{(x, g_1(x)) : x \in X\} = \{(x, g_2(x)) : x \in X\}$$

This does imply $g_1 = g_2$, so $F$ is faithful.

*(3) Fully faithfulness.* The map is not full, thus it is not fully faithful.

*(4) Conservativeness.* Suppose $F(f)$ is an isomorphism, meaning we can find $g \in \mathrm{Hom}_{\mathbf{Rel}}(Y, X)$ such that $F(f) \circ g = \Delta_{F(Y) \times F(Y)}$ and $g \circ F(f) = \Delta_{F(X) \times F(X)}$. Take the first condition:

$$\begin{aligned} F(f) \circ g &= \{(x, x') : \exists y \text{ such that } (x, y) \in F(f), (y, x') \in g\} \\ &= \{(x, x') : \exists y \text{ such that } (f(x), x') \in g\} \end{aligned}$$

Equating this to $\Delta_{F(X) \times F(X)}$ gives $x = x'$. Thus, $S$ is the graph of $g$ and thus the inverse of $f$. Thus $f$ is a bijection.

**Problem.** Prove that the category $\mathscr{C}$ is equivalent to the opposite category $\mathscr{C}^{\mathrm{op}}$ in the following cases:

1. $\mathscr{C}$ is the category of finite abelian groups ($\mathbf{Ab}^f$).
2. $\mathscr{C} = \mathbf{Rel}$.

*Part 2.* This part is surprisingly easy: given relation $f \subseteq X \times Y$, we map it to the relation $F(f) = \{(y, x) : (x, y) \in f\} \subseteq Y \times X$. It is straightforward to show that this map is fully faithful and essentially surjective.

*Part 1.* Define the functor $F : \mathbf{Ab}^f \to \mathbf{Ab}^f$ as follows:

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# 2 Limits

## 2.1 Basic Theory

In what follows, we would need the notion of *representable functors*.

**Definition 2.1** (Representable Functors)**.** Given functor $F : \mathscr{C}^{\mathrm{op}} \to \mathbf{Set}$, we call it **representable** if it is isomorphic to $h_{\mathscr{C}}(X)$ for some $X \in \mathscr{C}$ in $\mathscr{C}^{\wedge}$. We call $X$ the **representative** of $F$.
*Note:* Alternatively, we call $F$ representable if there exists representative $X \in \mathscr{C}$ such that $F(Y) \cong \mathrm{Hom}(Y, X)$ for every $Y \in \mathscr{C}$.

**Example.** Consider the forgetful functor $F : \mathbf{Vect}_k \to \mathbf{Set}$ that maps every vector space to the underlying set. We claim that $F$ is represented by field $k$. Indeed, we ought to show that $F(V) \cong \mathrm{Hom}(k, V)$ for each vector space $V$. Build the bijection as follows: $\eta_V : V \to \mathrm{Hom}(k, V)$ as $\eta_V(v) = \varphi_v$ where $\varphi_v(a) = a \cdot v$.

Now, we shift our focus towards products and coproducts.

**Definition 2.2** (Products and coproducts)**.** Let $\mathscr{C}$ be a category and consider the family $\{X_i\}_{i \in I}$ of objects in $\mathscr{C}$ indexed by a $\mathscr{U}$-small set $I$. Consider two functors:

$$\mathscr{C}^{\mathrm{op}} \to \mathbf{Set},\ Y \mapsto \prod_{i \in I} \mathrm{Hom}_{\mathscr{C}}(Y, X_i),$$
$$\mathscr{C} \to \mathbf{Set},\ Y \mapsto \prod_{i \in I} \mathrm{Hom}_{\mathscr{C}}(X_i, Y).$$

The representatives of these two functors are called **product** and **coproduct** and are denoted by $\prod_{i \in I} X_i$ and $\coprod_{i \in I} X_i$, respectively.

We shall now discuss the notion of limits.

**Definition 2.3.** Let $I$ and $\mathscr{C}$ be categories with $I$ small. A **projective system** (resp. an inductive system) in $\mathscr{C}$ indexed by $I$ is nothing but a functor $\beta : I^{\mathrm{op}} \to \mathscr{C}$ (resp. $\alpha : I \to \mathscr{C}$).

**Definition 2.4** (Projective limits in Set)**.** Assume $\mathscr{C} = \mathbf{Set}$ and consider the projective system $\beta : I^{\mathrm{op}} \to \mathbf{Set}$. The limit of $\beta$ is given by:

$$\lim \beta = \{(x_i)_i \in \prod_i \beta(i);\ \beta(s)(x_j) = x_i \text{ for all } s \in \mathrm{Hom}_I(i, j)\}.$$

**Definition 2.5** (Limits). Consider two functors $\beta : I^{\mathrm{op}} \to \mathscr{C}$ and $\alpha : I \to \mathscr{C}$. For $X \in \mathscr{C}$, define the following functors from $I^{\mathrm{op}}$ to **Set**:

$$\mathrm{Hom}_{\mathscr{C}}(X, \beta) : i \mapsto \mathrm{Hom}_{\mathscr{C}}(X, \beta(i)) \in \mathbf{Set},$$
$$\mathrm{Hom}_{\mathscr{C}}(\alpha, X) : i \mapsto \mathrm{Hom}_{\mathscr{C}}(\alpha(i), X) \in \mathbf{Set}$$

Assume that functor $X \mapsto \lim \mathrm{Hom}_{\mathscr{C}}(X, \beta)$ is representable. Then, we call it $\lim \beta$. In other words, for any $X \in \mathscr{C}$ we have the following isomorphism:

$$\mathrm{Hom}_{\mathscr{C}}(X, \lim \beta) \cong \lim \mathrm{Hom}_{\mathscr{C}}(X, \beta).$$

Similarly, for colimits we have:

$$\mathrm{Hom}_{\mathscr{C}}(\operatorname{colim} \alpha, X) \cong \lim \mathrm{Hom}_{\mathscr{C}}(\alpha, X).$$

## 2.2 Exercises

**Problem.** Let $I$ be a small set and $(X_i)_{i \in I}$ a family of sets indexed by $I$. Show that $\coprod_i X_i = \bigsqcup_i X_i$, the disjoint union of the sets $X_i$.

**Solution.** We should show that $Z := \bigsqcup_i X_i$ is the representative of the functor **Set** $\to$ **Set** given by the mapping $F(Y) = \prod_{i \in I} \mathrm{Hom}_{\mathscr{C}}(X_i, Y)$. More specifically, we need to show the following isomorphism: $F(Y) \cong \mathrm{Hom}(Z, Y)$ for every $Y \in \mathbf{Set}$. This unpacks into the following expression:

$$\prod_{i \in I} \mathrm{Hom}_{\mathscr{C}}(X_i, Y) \cong \mathrm{Hom}\left( \bigsqcup_{i \in I} X_i, Y \right)$$

We build the isomorphism $\Phi : \prod_{i \in I} \mathrm{Hom}_{\mathscr{C}}(X_i, Y) \to \mathrm{Hom}\left( \bigsqcup_{i \in I} X_i, Y \right)$ as follows:

$$\Phi((f_i)_{i \in I}) = G, \quad G(i, x) := f_i(x), \quad (i, x) \in \bigsqcup_{i \in I} X_i$$

**Inverse.** For a given $G$, define a family $\Psi(G) = (f_i)_{i \in I}$ by:

$$f_i := G \circ \iota_i : X_i \xrightarrow{\iota_i} Z \xrightarrow{G} Y, \quad \iota_i(x) := (i, x)$$

It is straightforward to show $\Psi \circ \Phi$ and $\Phi \circ \Psi$ yield identities.

**Problem.** Construct a natural map $\bigsqcup_{i \in I} \mathrm{Hom}_{\mathbf{Set}}(Y, X_i) \to \mathrm{Hom}_{\mathbf{Set}}(Y, \bigsqcup_{i \in I} X_i)$ and prove it is injective and not surjective in general.

**Solution.** Natural map $F$ is given as follows:

$$F : (i, f) \mapsto G : \mathbf{Set} \to \mathbf{Set} : Y \mapsto (i, f(Y))$$

In other words, given $\iota_i(x) = (i, x)$, we define $F(i, f) := \iota_i \circ f$.

**Problem.** Let $X, Y \in \mathscr{C}$ and consider the category $\mathscr{D}$ whose objects are triplets $Z \in \mathscr{C}$, $f : Z \to X$, $g : Z \to Y$, the morphisms being the natural ones. Prove that this category admits a terminal object if and only if the product $X \times Y$ exists in $\mathscr{C}$, and that in such a case this terminal object is isomorphic to $X \times Y, X \times Y \to X, X \times Y \to Y$. Deduce that if $X \times Y$ exists, it is unique up to unique isomorphism.

**Problem.** Let $I$ and $\mathscr{C}$ be two categories with $I$ small and denote by $\Delta$ the functor from $\mathscr{C}$ to $\mathrm{Fct}(I, \mathscr{C})$, which, to $X \in \mathscr{C}$, associates the constant functor $\Delta(X) : i \mapsto X$, $(i \to j) \mapsto \mathrm{id}_X$.

1. Assume that colimits indexed by $I$ exist. Prove the formula, for $\alpha : I \to \mathscr{C}$ and $Y \in \mathscr{C}$:
$$\mathrm{Hom}_{\mathscr{C}}(\mathrm{colim}\,\alpha, Y) \cong \mathrm{Hom}_{\mathrm{Fct}(I,\mathscr{C})}(\alpha, \Delta(Y)).$$
2. Assuming that limits exist, deduce the formula for $\beta : I^{\mathrm{op}} \to \mathscr{C}$ and $X \in \mathscr{C}$:
$$\mathrm{Hom}_{\mathscr{C}}(X, \lim \beta) \cong \mathrm{Hom}_{\mathrm{Fct}(I^{\mathrm{op}},\mathscr{C})}(\Delta(X), \beta)$$

*Part 1.* We shall build the isomorphism:

$$\Phi : \mathrm{Hom}_{\mathscr{C}}(\mathrm{colim}\,\alpha, Y) \to \mathrm{Hom}_{\mathrm{Fct}(I,\mathscr{C})}(\alpha, \Delta(Y)).$$

Take any morphism $f : \mathrm{colim}\,\alpha \to Y$. We need to associate it with some natural transformation $\theta : \alpha \to \Delta(Y)$. Let $\varepsilon_i : \alpha(i) \to \mathrm{colim}\,\alpha$ be a natural map. Then, we need to build a family $\theta_i : \alpha(i) \to \Delta(Y)(i)$. However note that $\Delta(Y)(i) = Y$, so we build $\theta_i : \alpha(i) \to Y$. So define $\theta_i$ as follows: $\theta_i = f \circ \varepsilon_i$. We now show that this is indeed the natural transformation. Consider the following commutative diagram:

$$\begin{array}{ccc} \alpha(i) & \xrightarrow{\theta_i = f\circ\varepsilon_i} & \Delta(Y)(i) = Y \\ {\scriptstyle \alpha(f)}\downarrow & & \downarrow{\scriptstyle (\Delta Y)(f)} \\ \alpha(j) & \xrightarrow{\theta_j = f\circ\varepsilon_j} & \Delta(Y)(j) = Y \end{array}$$

*Part 2.* We do similarly. Define the isomorphism:

$$\Psi : \mathrm{Hom}_{\mathscr{C}}(X, \lim \beta) \to \mathrm{Hom}_{\mathrm{Fct}(I^{\mathrm{op}},\mathscr{C})}(\Delta(X), \beta)$$

Take any morphism $f : X \to \lim \beta$. For this $f$ we shall build the natural transformation $\alpha : \Delta(X) \to \beta$. For that I take any $i$ and build $\alpha(i) : \Delta(X)(i) \to \beta(i)$ which basically gives me $\alpha(i) : X \to \beta(i)$. Now suppose $\pi_i : \lim \beta \to X_i$ is the natural morphism. Take $\alpha(i) := \pi_i \circ f$.

**Problem.** (i) Let $\mathscr{C}$ be a category which admits colimits indexed by a category $I$. Let $\alpha : I \to \mathscr{C}$ be a functor and let $X \in \mathscr{C}$. Construct the natural morphism

$$\mathrm{colim}_i \mathrm{Hom}_{\mathscr{C}}(X, \alpha(i)) \to \mathrm{Hom}_{\mathscr{C}}(X, \mathrm{colim}_i \alpha(i)).$$

(ii) Let $\mathbf{k}$ be a field and denote by $\mathbf{k}[x]^{\leq n}$ the $\mathbf{k}$-vector space consisting of polynomials of degree $\leq n$. Prove the isomorphism $\mathbf{k}[x] \cong \mathrm{colim}_n \mathbf{k}[x]^{\leq n}$ in $\mathrm{Mod}(\mathbf{k})$ and, noticing that $\mathrm{id}_{\mathbf{k}[x]} \notin \mathrm{colim}_n \mathrm{Hom}_{\mathbf{k}}(\mathbf{k}[x], \mathbf{k}[x]^{\leq n})$, deduce that the morphism

**Problem.** We consider the ordered set $\mathbb{N}$ as a category. Hence, for a category $\mathscr{C}$, a functor $\alpha : \mathbb{N} \to \mathscr{C}$ is defined by the data of the objects $\alpha(n) \in \mathscr{C}$, $n \in \mathbb{N}$, and the morphisms $\alpha(n < n+1) : \alpha(n) \to \alpha(n+1)$.
(i) Consider the functor $\alpha : \mathbb{N} \to \mathrm{Mod}(\mathbb{Z})$ given by $\alpha(n) = \mathbb{Z}$ and $\alpha(n < n+1) = 2\cdot : \mathbb{Z} \to \mathbb{Z}$. Calculate $\operatorname{colim} \alpha$.
(ii) Give an example of a functor $\alpha : \mathbb{N} \to \mathrm{Mod}(\mathbb{Z})$ in which all $\alpha(n)$ are not 0 and all morphisms $\alpha(n < n+1)$ are not 0 but $\operatorname{colim}(\alpha) \cong 0$.

Part (i). This is a directed colimit, so we can define the relation over $\bigsqcup_{i\in I} \mathbb{Z} = \mathbb{Z}^{\sqcup}$ as follows:

$$(x,i)\mathscr{R}(y,j) \iff \exists i,j,k \in \mathbb{N} : 2^{k-i}x = 2^{k-j}y \iff \frac{x}{2^i} = \frac{x}{2^j}$$

So $\mathbb{Z}^{\sqcup}/\mathscr{R} \cong \{\frac{x}{2^n} : x \in \mathbb{Z}, n \in \mathbb{N}\} = \boxed{\mathbb{Z}[1/2]}$.

Part (ii). This happens if $\mathscr{R} = \bigsqcup_{i\in\mathbb{N}} \alpha(i) \times \bigsqcup_{i\in\mathbb{N}} \alpha(i)$. This can happen, for instance, in the following case: suppose $\alpha(n)$ is the group of rational points on the unit circle: say, $\mathbb{Q}/\mathbb{Z}$. Define the morphism $\alpha(n < n+1) = \star^{n+1}$. Now note that every element is equivalent to 1. Indeed:

# 3 Additive Categories

**Problem.** Let $\mathscr{C}$ be an additive category and let $X \in C(\mathscr{C})$ with differential $d_X$. Let $\{a_n\}_{n\in\mathbb{Z}}$ be a sequence in $\mathbb{Z}$. Define the morphism $\delta_X : X \to X[1]$ by setting $\delta_X^n = a_n d_X^n$. Prove that $\delta_X$ is a morphism in $C(\mathscr{C})$ and is homotopic to zero.

**Proof.** Recall that $f : X \to Y$ is a morphism if the following diagram commutes:

$$\begin{array}{ccccccc} \dots & \longrightarrow & X^n & \xrightarrow{d_X^n} & X^{n+1} & \longrightarrow & \dots \\ & & \downarrow{\scriptstyle f^n} & & \downarrow{\scriptstyle f^{n+1}} & & \\ \dots & \longrightarrow & Y^n & \xrightarrow{d_Y^n} & Y^{n+1} & \longrightarrow & \dots \end{array}$$

In our particular case, as $\delta : X \to X[1]$, the diagram looks as follows:

$$\begin{array}{ccccccc} \dots & \longrightarrow & X^n & \xrightarrow{d_X^n} & X^{n+1} & \longrightarrow & \dots \\ & & \downarrow{\scriptstyle \delta^n} & & \downarrow{\scriptstyle \delta^{n+1}} & & \\ \dots & \longrightarrow & X^{n+1} & \xrightarrow{d_{X[1]}^n} & X^{n+2} & \longrightarrow & \dots \end{array}$$

So we are left to prove that:

$$\delta^{n+1} \circ d_X^n = d_{X[1]}^n \circ \delta^n$$

Now by expanding $\delta^n$ and using the fact that $d_{X[1]}^n = -d_X^{n+1}$:

$$\text{LHS} = a_{n+1} d_X^{n+1} \circ d_X^n, \quad \text{RHS} = -d_X^{n+1} \circ a_n d_X^n$$

Trivially, both sides are 0, so our condition is met.

Now we prove that $\delta_X$ is homotopic to zero. By definition, if $f$ is homotopic to zero if for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ such that:

$$f^p = s^{p+1} \circ d_X^p + d_Y^{p-1} \circ s^p$$

Now, in our terms, we have: for all $p \in \mathbb{Z}$ we should find an endomorphism $s^p : X^p \to X^p$ such that:

$$a_p d_X^p = s^{p+1} \circ d_X^p - d_X^p \circ s^p$$

Define $s^p := b_p \cdot \mathrm{id}_{X^p}$ for some $\{b_p\}_{p\in\mathbb{Z}}$. By substituting, we obtain:

$$a_p d_X^p = b_{p+1}\mathrm{id}_{X^p} \circ d_X^p - d_X^p \circ b_p \mathrm{id}_{X^p} = b_{p+1} d_X^p - b_p d_X^p = (b_{p+1} - b_p) d_X^p$$

Thus we require $b_{p+1} - b_p = a_p$ for all $p$. Thus, for instance, define $b_0 := 0$. The resultant morphism is thus $s^p = b_p \mathrm{id}_{X^p}$.

**Problem.** Let $\mathscr{C}$ be an additive category, $f, g : X \rightrightarrows Y$ two morphisms in $C(\mathscr{C})$. Prove that $f$ and $g$ are homotopic if and only if there exists a commutative diagram in $C(\mathscr{C})$.

$$\begin{array}{ccccc} Y & \xrightarrow{\alpha(f)} & \mathrm{Mc}(f) & \xrightarrow{\beta(f)} & X[1] \\ \| & & \downarrow u & & \| \\ Y & \xrightarrow{\alpha(g)} & \mathrm{Mc}(g) & \xrightarrow{\beta(g)} & X[1] \end{array}$$

Moreover, assert that in such case $u$ is an isomorphism in $C(\mathscr{C})$.

**Solution.** First, assume $f$ and $g$ are homotopic. This means that there exists a map $s^p$ s.t.:

$$f^p - g^p = s^{p+1} \circ d_X^p + d_Y^{p-1} \circ s^p$$

Now we shall construct the mapping $u : \mathrm{Mc}(f) \to \mathrm{Mc}(g)$. The map is defined as follows:

$$u^p : X^{p+1} \oplus Y^p \to X^{p+1} \oplus Y^p, \quad u^p(x,y) = \begin{bmatrix} u_{1,1}^p & u_{1,2}^p \\ u_{2,1}^p & u_{2,2}^p \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u_{1,1}^p(x) + u_{1,2}^p(y) \\ u_{2,1}^p(x) + u_{2,2}^p(y) \end{bmatrix}$$

What do we require from such a map? We need for diagram to commute, so we want $u \circ \alpha(f) = \alpha(g)$ and $\beta(g) \circ u = \beta(f)$. First condition means:

$$u^p \circ \alpha^p(f)(y) = u^p(0,y) = (u_{1,2}^p(y), u_{2,2}^p(y)) = \beta^p(f)(y) = (0,y)$$

Thus we have $u_{1,2}^p = 0$ and $u_{2,2}^p = \mathrm{id}_{Y^p}$. Second condition implies:

$$\beta^p(g) \circ u(x,y) = \beta^p(g) \begin{bmatrix} u_{1,1}^p(x) + u_{1,2}^p(y) \\ u_{2,1}^p(x) + u_{2,2}^p(y) \end{bmatrix} = u_{1,1}^p(x) + u_{1,2}^p(y) = x$$

Note however that $u_{1,2}^p(y) = 0$, so $u_{1,1}^p(x) = x$. Thus, we take $u_{1,1}^p = \mathrm{id}_{X^{p+1}}$. Thus, we have:

$$u^p \star = \begin{bmatrix} \mathrm{id}_{X^{p+1}} & 0 \\ \gamma^p & \mathrm{id}_{Y^p} \end{bmatrix} \star$$

So we ought to find $\gamma^p$. For that, we want $u$ to be a chain map:

$$d_{\mathrm{Mc}(g)}^n \circ u^n = u^{n+1} \circ d_{\mathrm{Mc}(f)}^n$$

$$\begin{array}{ccccccc} \ldots & \longrightarrow & X^n \oplus Y^{n-1} & \xrightarrow{d^n_{\mathrm{Mc}(f)}} & X^{n+1} \oplus Y^n & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle u^n} & & \downarrow{\scriptstyle u^{n+1}} & & \\ \ldots & \longrightarrow & X^n \oplus Y^{n-1} & \xrightarrow{d^n_{\mathrm{Mc}(g)}} & X^{n+1} \oplus Y^n & \longrightarrow & \ldots \end{array}$$

By expanding, we have:

$$\begin{bmatrix} -d_X^{n+1} & 0 \\ g^{n+1} & d_Y^n \end{bmatrix} \begin{bmatrix} \mathrm{id}_{X^{n+1}} & 0 \\ \gamma^n & \mathrm{id}_{Y^n} \end{bmatrix} = \begin{bmatrix} \mathrm{id}_{X^{n+2}} & 0 \\ \gamma^{n+1} & \mathrm{id}_{Y^{n+1}} \end{bmatrix} \begin{bmatrix} -d_X^{n+1} & 0 \\ f^{n+1} & d_Y^n \end{bmatrix}$$

By simplifying both sides, we arrive at:

$$\begin{bmatrix} -d_X^{n+1} & 0 \\ g^{n+1} + d_Y^n \circ g^n & d_Y^n \end{bmatrix} = \begin{bmatrix} -d_X^{n+1} & 0 \\ -\gamma^{n+1} \circ d_X^{n+1} + f^{n+1} & d_Y^n \end{bmatrix}$$

As can be seen, the nly requirement we impose:

$$g^{n+1} + d_Y^n \circ g^n = -\gamma^{n+1} \circ d_X^{n+1} + f^{n+1}$$

Or, by rearranging:

$$f^{n+1} - g^{n+1} = \gamma^{n+1} \circ d_X^{n+1} + d_Y^n \circ g^n$$

Thus, simply set $\gamma^n := s^{n+1}$. All in all:

$$u^p \star = \begin{bmatrix} \mathrm{id}_{X^{p+1}} & 0 \\ s^{p+1} & \mathrm{id}_{Y^p} \end{bmatrix} \star$$

The opposite direction is essentially the same. We start with arbitrary $u^p$ and from the same derivations, arrive that $u^p$ must have this specific form. The reason why is that it has the following inverse map:

$$v^p := \begin{bmatrix} \mathrm{id}_{X^{p+1}} & 0 \\ -s^{p+1} & \mathrm{id}_{Y^p} \end{bmatrix}$$

**Problem.** A category with translation $(\mathscr{A}, T)$ is a category $\mathscr{A}$ together with an equivalence $T : \mathscr{A} \to \mathscr{A}$. A differential object $(X, d_X)$ in a category with translation $(\mathscr{A}, T)$ is an object $X \in \mathscr{A}$ together with a morphism $d_X : X \to T(X)$. A morphism $f : (X, d_X) \to (Y, d_Y)$ of differential objects is a commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{d_X} & TX \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle T(f)} \\ Y & \xrightarrow{d_Y} & TY \end{array}$$

One denotes by $\mathscr{A}_d$ the category consisting of differential objects and morphisms of such objects. If $\mathscr{A}$ is additive, one says that a differential object $(X, d_X)$ in $(\mathscr{A}, T)$ is a complex if the composition $X \xrightarrow{d_X} T(X) \xrightarrow{T(d_X)} T^2(X)$ is zero. One denotes by $\mathscr{A}_c$ the full subcategory of $\mathscr{A}_d$ consisting of complexes.

1. Let $\mathscr{C}$ be a category. Denote by $\mathbb{Z}_d$ the set $\mathbb{Z}$ considered as a discrete category and still denote $\mathbb{Z}$ as the ordered set $(\mathbb{Z}, \leq)$ considered as a category. Prove that $\mathrm{Fct}(\mathbb{Z}_d, \mathscr{C})$ is a category with translation.
2. Show that the category $\mathrm{Fct}(\mathbb{Z}, \mathscr{C})$ may be identified to the category of differential objects in $\mathrm{Fct}(\mathbb{Z}_d, \mathscr{C})$.
3. Let $\mathscr{C}$ be an additive category. Show that the notions of differential objects and complexes given above coincide with those in Definition 4.2.1 when choosing $\mathscr{A} = C(\mathscr{C})$ and $T = [1]$.

**Solution.**

*Part (i).* First note that $\mathrm{Fct}(\mathbb{Z}_d, \mathscr{C})$ is nothing but a category of sequences $\{X_i\}_{i \in \mathbb{Z}}, X_i = \alpha(i)$ for given $\alpha : \mathbb{Z}_d \to \mathscr{C}$, equipped with endomorphisms $X_i \to X_i$. To show that it is a category with translation, we build an equivalence $T : \mathrm{Fct}(\mathbb{Z}_d, \mathscr{C}) \to \mathrm{Fct}(\mathbb{Z}_d, \mathscr{C})$. We build the following family of translations:

$$T^p : X_i \mapsto X_{i+p}, X_i \to X_i \mapsto X_{i+p} \to X_{i+p}$$

Inverse map is given by $T^{-p}$.

*Part (ii).* We consider how differential objects in $\mathrm{Fct}(\mathbb{Z}_d, \mathscr{C})$ look like under translation $T^1$. The tuple $(X, d_X)$ is essentially a sequence $\{X_i\}_i$ equipped with $d_X : \{X_i\}_i \to \{X_{i+1}\}_i$. Moreover, since $d_X$ is a morphism, the following diagram commutes:

$$\begin{array}{ccc} X_i & \xrightarrow{d_X^i} & X_{i+1} \\ \downarrow{\scriptstyle f_i} & & \downarrow{\scriptstyle f_{i+1}} \\ Y_i & \xrightarrow{d_Y^i} & Y_{i+1} \end{array}$$

That said, category of differential objects is exactly $\mathrm{Fct}(\mathbb{Z}, \mathscr{C})$:

$$\mathrm{Fct}(\mathbb{Z}, \mathscr{C}) \cong \mathrm{DiffObjs}(\mathrm{Fct}(\mathbb{Z}_d, \mathscr{C}), T)$$

*Part (iii).* Let us see what condition of composition $X \xrightarrow{d_X} T(X) \xrightarrow{T(d_X)} T^2(X)$ being zero using $T = [1]$ and $\mathscr{A} = C(\mathscr{C})$ means.

One has $X \in C(\mathscr{C})$, so $X$ is a complex. In turn, $d_X$ is nothing but $d_X : X \to X[1]$, so $d_X^n : X^n \to X^{n+1}$. Similarly, $T(d_X^n) = -d_X^{n+1}$. So we translate the diagram to:

$$X^n \xrightarrow{d_X^n} X^{n+1} \xrightarrow{-d_X^{n+1}} X^{n+2}$$

Stating that $d_X^n \circ d_X^{n+1} = 0$ is equivalent to saying that this composition is $0$ for all $n$.

# 4 Abelian Categories

**Problem.** Consider two complexes in an abelian category $\mathscr{C}$:

$$X_1' \to X_1 \to X_1'', \quad X_2' \to X_2 \to X_2''.$$

Prove that they are exact iff the sequence

$$X_1' \oplus X_2' \to X_1 \oplus X_2 \to X_1'' \oplus X_2''$$

is exact.

**Solution.**

**Problem.** If $M$ is a $\mathbb{Z}$-module, set $M^\vee := \mathrm{Hom}(M, \mathbb{Q}/\mathbb{Z})$.

1. Prove that $\mathbb{Q}/\mathbb{Z}$ is injective in $\mathrm{Mod}(\mathbb{Z})$.
2. Prove that the map $\mathrm{Hom}_{\mathbb{Z}}(M, N) \to \mathrm{Hom}_{\mathbb{Z}'}(M^\vee, N^\vee)$ is injective for $N, M \in \mathrm{Mod}(\mathbb{Z})$.
3. Prove that if $P$ is a right projective $A$-module, then $P^\vee$ is left $A$-injective.
4. Let $M$ be an $A$-module. Prove that there exists an injective $A$-module $I$ and a monomorphism $M \to I$.

**Solution.**

*Part (i).* By Proposition 5.3.2, object $\mathbb{Q}/\mathbb{Z} \in \mathrm{Mod}(\mathbb{Z})$ is injective if and only if, the following diagram commutes in $\mathscr{C}$ for $X', X \in \mathrm{Mod}(\mathbb{Z})$ and the row is exact:

$$\begin{array}{ccccc} 0 & \longrightarrow & X' & \xrightarrow{f} & X \\ & & \downarrow{\scriptstyle k} & \swarrow{\scriptstyle h} & \\ & & \mathbb{Q}/\mathbb{Z} & & \end{array}$$

This way, $f$ is an injective homomorphism between abelian groups $X'$ and $X$, while $k : X' \to \mathbb{Q}/\mathbb{Z}$ is a character of $X'$. We shall show that $h : X' \to \mathbb{Q}/\mathbb{Z}$ is determined uniquely by $f$ and $k$. In fact, since $k = h \circ f$, we find $h = k \circ f^{-1}$.

# 5 Sheaves on sites

**Problem** (Exercise 1.3)**.** Assume $X$ is a topological space and $U \in \mathrm{Op}_X$. Prove that the composition of morphisms of presites $U \xrightarrow{i_U} X \xrightarrow{j_U} U$ is isomorphic to $\mathrm{id}_U$.

**Proof.** We shall show that composition $\mathrm{Op}_U \xrightarrow{j_U} \mathrm{Op}_X \xrightarrow{i_U} \mathrm{Op}_U$ gives an identity morphism $\mathrm{id}_{\mathrm{Op}_U}$. Let us compute $i_U \circ j_U(V)$ for some $V \in \mathrm{Op}_U$. Note that $j_U(V) = V$ — we simply include $V$ to the larger category $\mathrm{Op}_X$. Then, we apply $i_U$ and get $i_U(V) = U \cap V$. However, since $V \in \mathrm{Op}_U$, in particular one has $V \subseteq U$, so $U \cap V = V$. Therefore, $i_U \circ j_U(V) = V$ for any $V \in \mathrm{Op}_U$, thus $i_U \circ j_U \cong \mathrm{id}_{\mathrm{Op}_U}$.

**Problem** (Exercise 1.8)**.** Let $X$ be a topological space and let $X_{\text{finite}}$ be the site defined as follows: $\mathscr{S} \subset \mathrm{Op}_U$ is a covering of $U$ if there exists a finite subset $\mathscr{S}' \subset \mathscr{S}$ such that $\bigcup_{V \in \mathscr{S}'} V = U$. Denote by $f : X \to X_{\text{finite}}$ the natural morphism. Prove that the presheaf $\mathscr{C}_X^{0,b}$ of $\mathbb{C}$-valued continuous bounded functions is a sheaf on the site $X_{\text{finite}}$ and calculate $f^{-1}\mathscr{C}_X^{0,b}$.

**Solution.** We first show that the presheaf $\mathscr{C}_X^{0,b}$ is a sheaf. We need to check two properties:

> *(i)* For any open subset $U \subseteq X$, any open finite covering $\{U_i\}_{i \in I}$, any continuous bounded $s \in F(U)$ satisfying $s|_{U_i} = 0$ for all $i$, one has $s = 0$.

This property is obvious. Indeed, if $s$ is zero on each $U_i$, it is clearly zero on $U$. The second property is less trivial:

> *(ii)* For any open subset $U \subseteq X$, any open finite covering $\{U_i\}_{i \in I}$ and family $\{s_i \in F(U_i)\}_{i \in I}$ satisfying $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j$, there exists $s \in F(U)$ with $s|_{U_i} = s_i$.

For each $x \in U$, pick $x \in U_i$ and set $s(x) = s_i(x)$. It is well-defined since if $x \in U_j$, $j \neq i$, then $s_i(x) = s_j(x)$ since $x \in U_i \cap U_j$ and $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. We shall check both continuity and boundness. Former is trivial since all $s_i$ are continuous. For boundness, notice that $|s_i(x)| \leq M_i$ for $x \in U_i$. Since $I$ is finite, define $M := \max_{i \in I} M_i$. Then, $|s(x)| \leq M$. Thus $s$ is bounded.

We shall now find $f^{-1}\mathscr{C}_X^{0,b}$. By definition,

$$f^{-1}F : U \mapsto \mathrm{colim}_{V \subseteq U, V \in X_{\text{finite}}} F(V) \cong \bigsqcup_{V \subseteq U, V \in X_{\text{finite}}} F(V) \quad / \quad \mathscr{R}$$

with $s_1 \mathscr{R} s_2$ for $s_1 \in F(V_1)$, $s_2 \in F(V_2)$ iff exists $V_3 \subseteq V_1 \cap V_2$, $V_3 \in X_{\text{finite}}$ such that $s_1|_{V_3} = s_2|_{V_3}$.

Let us see what is generated by $[(s, V)]$ for bounded continuous $s : V \to \mathbb{C}$. Another function $t : W \to \mathbb{C}$ is in $[(s, V)]$ iff $s$ and $t$ agree on some subset of $V \cap W$. In paritcular, this means that $(s|_W, W) \in [(s, V)]$ for $W \subseteq V$. However, this implies that in fact $[(s, V)] \cong \varphi$ where $\varphi$ is some continuous but not necessarily bounded function. Thus, $\boxed{f^{-1}\mathscr{C}_X^{0,b} \cong \mathscr{C}_X^0}$.

# 6 Abelian Sheaves

**Problem** (Exercise 1.7)**.** Let $X = \mathbb{R}^2$ with coordinates $(x, y)$, $Y = \mathbb{R}$ and let $\pi : X \to Y$ be the projection $(x, y) \mapsto x$. Let

$$Z_0 = \{(x, y) \in \mathbb{R}^2 : xy > 1\}, \quad Z_1 = \{(x, y) \in \mathbb{R}^2 : xy \geq 1, x > 0\},$$
$$Z_2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Calculate $R\pi_* \mathbf{k}_{XZ_i}$ for $i \in \{0, 1, 2\}$.

**Solution.** We call the following isomorphism:

$$(R^p \pi_* F)_x \cong H^p(\pi^{-1}(x), F|_{\pi^{-1}(x)})$$

**Case for** $Z_0$**.** We have $F|_{\pi^{-1}(x)} \cong \mathbb{R}$